

Chapter 7

System and Memory

7.1 Overview

ESP32-P4 integrates two processors:

- a high-performance 32-bit RISC-V dual-core processor (HP CPU), five-stage pipeline, clock frequency up to 400 MHz.
- a low-power 32-bit RISC-V single-core processor (LP CPU), two-stage pipeline, clock frequency up to 40 MHz.

All internal memory, external memory, and peripherals are located on the HP CPU and LP CPU buses.

7.2 Features

- **Address Space**

- 896 KB of HP internal memory address space, accessible by the instruction bus or data bus of HP CPU via cache
- 8 KB of HP internal memory address space, accessible by the instruction bus or data bus of HP CPU
- 48 KB of LP internal memory address space, accessible by the instruction bus or data bus of LP CPU with zero latency
- 1256 KB of peripheral address space
- 64 MB of external flash virtual address space, accessible by the instruction bus or data bus
- 64 MB of external RAM virtual address space, accessible by the instruction bus or data bus
- 768 KB of internal DMA address space
- 128 MB of external DMA address space

- **Internal Memory**

- 128 KB of HP ROM
- 768 KB of HP L2MEM
- 8 KB of HP SPM (Scratchpad Memory)
- 16 KB of LP ROM
- 32 KB of LP SRAM

- **External Memory**

- Supports up to 64 MB of external flash and RAM
- **Peripheral Space**
 - 96 modules/peripherals in total
- **GDMA**
 - 7 GDMA-AHB supported modules/peripherals
 - 6 GDMA-AXI supported modules/peripherals

7.3 Functional Description

7.3.1 Address Mapping

Figure 7.3-1 illustrates the system structure and address mapping. All the non-reserved addresses are accessible via both the instruction bus and the data bus, meaning that the instruction bus and the data bus share the same address space.

Both the data bus and instruction bus of the HP CPU and LP CPU are little-endian. However, the HP CPU's data bus (DBUS) has a 128-bit data width, while other buses maintain a 32-bit data width.

The HP CPU can access data via the data bus using single-byte, double-byte, 4-byte alignment, and, in case of AI Instruction, up to 16-byte alignment. The LP CPU can access data via the data bus using single-byte, double-byte, and 4-byte alignment.

The HP CPU has the following access capabilities:

- Direct access to HP SPM via both the data bus and instruction bus.
- Access to internal memory and external memory mapped into the address space via cache, including:
 - 128 KB HP ROM (0x4FC0_0000 ~ 0x4FC1_FFFF)
 - 768 KB HP L2MEM (0x4FF0_0000 ~ 0x4FFB_FFFF)
 - 64 MB external flash (0x4000_0000 ~ 0x43FF_FFFF)
 - 64 MB external RAM (0x4800_0000 ~ 0x4BFF_FFFF)

Note:

Software can use the addresses starting with "0x4" in two ways: cache-able access or noncache-able access, depending on the control of CPU Physical Memory Attribution (PMA). Cache-able access is done via cache. Noncache-able access is done without cache, which is similar to the direct access to the addresses starting with "0x8", however, such direct access to "0x8" addresses is usually for debugging.

- Direct access to internal memory and external memory mapped into the following address space without cache. This access is slower than that via cache.
 - 128 KB HP ROM (0x8FC0_0000 ~ 0x8FC1_FFFF)
 - 768 KB HP L2MEM (0x8FF0_0000 ~ 0x8FFB_FFFF)
 - 32 KB LP SRAM (0x5010_8000 ~ 0x5010_FFFF)

- 64 MB external flash (0x8000_0000 ~ 0x83FF_FFFF)
- 64 MB external RAM (0x8800_0000 ~ 0x8BFF_FFFF)
- Direct access to modules/peripherals via the data bus, including:
 - HP CPU Peripherals
 - HP Peripherals
 - LP Peripherals

The LP CPU's access capabilities include:

- Direct access to LP SRAM and LP ROM via both the data bus and instruction bus.
- Access to internal memory and external memory, including:
 - 128 KB HP ROM (0x4FC0_0000 ~ 0x4FC1_FFFF)
 - 768 KB HP L2MEM (0x4FF0_0000 ~ 0x4FFB_FFFF)
 - 64 MB external flash (0x4000_0000 ~ 0x43FF_FFFF)
 - 64 MB external RAM (0x4800_0000 ~ 0x4BFF_FFFF)
- Direct access to modules/peripherals via the data bus, including:
 - HP CPU Peripherals
 - HP Peripherals
 - LP Peripherals

Table 7.3-1 – cont'd from previous page

Bus Type	Boundary Address		Size	Target *
	Low Address	High Address		
Data/Instruction bus	0x3010_0000	0x3010_1FFF	8 KB	HP SPM
	0x3010_2000	0x3FEF_FFFF		Reserved
Data/Instruction bus	0x3FF0_0000	0x3FF1_FFFF	128 KB	HP CPU peripherals
	0x3FF2_0000	0x3FFF_FFFF		Reserved
Data/Instruction bus	0x4000_0000	0x43FF_FFFF	64 MB	External flash
	0x4400_0000	0x47FF_FFFF		Reserved
Data/Instruction bus	0x4800_0000	0x4BFF_FFFF	64 MB	External RAM
	0x4C00_0000	0x4FBF_FFFF		Reserved
Data/Instruction bus	0x4FC0_0000	0x4FC1_FFFF	128 KB	HP ROM
	0x4FC2_0000	0x4FEF_FFFF		Reserved
Data/Instruction bus	0x4FF0_0000	0x4FFB_FFFF	768 KB	HP L2MEM
	0x4FFC_0000	0x4FFF_FFFF		Reserved
Data/Instruction bus	0x5000_0000	0x500F_FFFF	1 MB	HP peripherals
Data/Instruction bus	0x5010_0000	0x5010_3FFF	16 KB	LP ROM
	0x5010_4000	0x5010_7FFF		Reserved
Data/Instruction bus	0x5010_8000	0x5010_FFFF	32 KB	LP SRAM
Data/Instruction bus	0x5011_0000	0x5012_FFFF	128 KB	LP peripherals
	0x5013_0000	0x7FFF_FFFF		Reserved
Data/Instruction bus	0x8000_0000	0x83FF_FFFF	64 MB	External flash (direct access by CPU)
	0x8400_0000	0x87FF_FFFF		Reserved
Data/Instruction bus	0x8800_0000	0x8BFF_FFFF	64 MB	External RAM (direct access by CPU)
	0x8C00_0000	0x8FBF_FFFF		Reserved
Data/Instruction bus	0x8FC0_0000	0x8FC1_FFFF	128 KB	HP ROM (direct access by CPU)
	0x8FC2_0000	0x8FEF_FFFF		Reserved
Data/Instruction bus	0x8FF0_0000	0x8FFB_FFFF	768 KB	HP L2MEM (direct access by CPU)
	0x8FFC_0000	0xFFFF_FFFF		Reserved

* Address starting with 0x4xxx_xxxx can be configured as noncache-able or cache-able, depending on CPU PMA, whereas address starting with 0x8xxx_xxxx is accessed directly by CPU.

* For the targets accessible by HP CPU or LP CPU, see Figure 7.3-1.

7.3.2 Internal Memory

ESP32-P4 has various types of internal memory:

- **HP ROM (128 KB):** This read-only memory is dedicated to the HP system and is not programmable. It contains the ROM code and read-only data of some low-level system software. The HP ROM is accessed by the HP CPU through the I2 cache at half the frequency of HP CPU.
- **HP L2MEM (768 KB):** A volatile memory accessible by the HP CPU, LP CPU, or DMA peripherals at half the frequency of the HP CPU. HP L2MEM can be configured to retain power during Light-sleep mode, making it suitable for data retention or register backup.
- **LP ROM (16 KB):** This read-only memory serves the LP system, containing the code for booting the LP CPU and some basic system functions. The LP ROM is accessed by the LP CPU with zero latency.

- **LP SRAM (32 KB):** A volatile memory accessible by both the HP CPU and LP CPU, operating at the same frequency as the LP CPU.
- **HP SPM (8 KB):** A volatile memory accessed by the HP CPU, finishing one access in two cycles.

1. Details for HP ROM

The 128 KB HP ROM is a read-only memory accessed by the HP CPU through the instruction bus or data bus via the following addresses as shown in Table 7.3-1:

- 0x4FC0_0000 (cache-able or noncache-able, configuration-dependent) ~ 0x4FC1_FFFF (cache-able or noncache-able, configuration-dependent).
- 0x8FC0_0000 (direct access by CPU) ~ 0x8FC1_FFFF (direct access by CPU).

2. Details for HP L2MEM

This 768 KB HP L2MEM is a read-and-write memory accessed by the HP CPU or LP CPU through the instruction bus or data bus, or by DMA master through AHB matrix or AXI matrix via the following addresses as shown in Table 7.3-1:

- 0x4FF0_0000 (cache-able or noncache-able, configuration-dependent) ~ 0x4FFB_FFFF (cache-able or noncache-able, configuration-dependent).
- 0x8FF0_0000 (direct access by CPU) ~ 0x8FFB_FFFF (direct access by CPU).

L2MEM supports multiple capacity split configurations: L2 RAM and L2 cache share a total capacity of 768 KB, and one of the following four configurations can be selected:

- Config 0: L2 RAM 256 KB, L2 cache 512 KB
- Config 1: L2 RAM 512 KB, L2 cache 256 KB
- Config 2: L2 RAM 640 KB, L2 cache 128 KB
- Config 3: L2 RAM 768 KB, L2 cache 0 KB

The address layout of HP L2MEM for the above four configurations is shown in Figure 7.3-2.

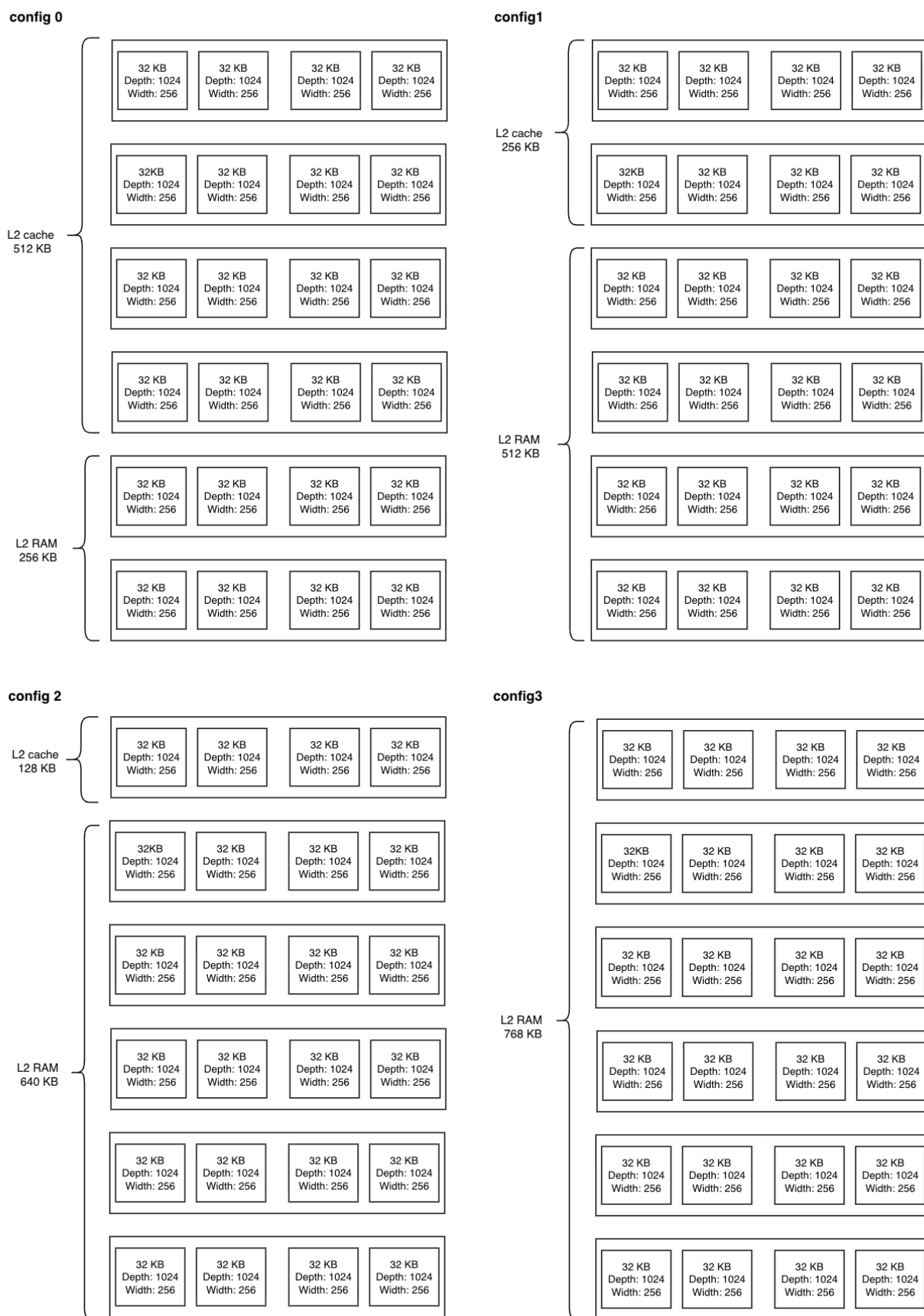


Figure 7.3-2. HP L2MEM Address Layout

HP L2MEM can be accessed by various peripherals:

- by USB 2.0 High-speed, USB 2.0 Full-speed, GMAC, SDMMC, TRACE, GDMA-AHB, and SYSMON via AHB matrix.
- by GDMA, GDMA-AXI, 2D-DMA, and H264 Encoder via AXI matrix.

The HP L2MEM supports Error Correction Code (ECC) check to correct one-bit memory flip error. In the event of a one-bit flip error, the data in HP L2MEM still remains valid. Before ECC check, HP L2MEM itself must be initialized first.

HP L2MEM consists of six memory units, each occupying 128 KB. The initialization process allows individual configuration of each unit for power considerations. It is recommended to initialize memory units sequentially rather than simultaneously. This sequential approach helps prevent excessive power consumption and avoids triggering a brownout condition when initializing the entire 768 KB HP L2MEM.

Detailed Configuration Steps:

Take unit0 as an example, other units follow the same configuration steps.

- Set `HP_SYSTEM_L2_MEM_UNIT0_REFRESH_EN` and clear `HP_SYSTEM_L2_MEM_REFRESH_CNT_RESET` to start initializing HP L2MEM.
- Read the value of `HP_SYSTEM_L2_MEM_UNIT0_REFRESH_DONE`.
- Once `HP_SYSTEM_L2_MEM_UNIT0_REFRESH_DONE` reaches a value of 1, set `HP_SYSTEM_L2_RAM_UNIT0_ECC_EN` to start ECC check.
- (Recommended) clear `HP_SYSTEM_L2_MEM_UNIT0_REFRESH_EN` and set `HP_SYSTEM_L2_MEM_REFRESH_CNT_RESET` after `HP_SYSTEM_L2_MEM_UNIT0_REFRESH_DONE` reaches 1 to complete the initialization process.

During the initialization process, avoid any access to HP L2MEM unit being initialized.

3. Details for LP ROM

This 16 KB LP ROM is a read-only memory accessed by LP CPU through the instruction bus or data bus via their shared address `0x5010_0000 ~ 0x5010_3FFF` as shown in Table 7.3-1.

4. Details for LP SRAM

This 32 KB LP SRAM is a read-and-write memory accessed by the HP CPU via AHB matrix and by the LP CPU through the instruction bus or data bus via their shared address `0x5010_8000 ~ 0x5010_FFFF` as shown in Table 7.3-1. LP SRAM supports atomic operation.

5. Details for HP SPM

This 8 KB HP SPM is a read-and-write memory accessed by the HP CPU through the instruction bus or data bus via their shared address `0x3010_0000 ~ 0x3010_1FFF` as shown in Table 7.3-1. Additionally, HP SPM supports atomic operation of the HP CPU.

HP SPM supports parity check to detect one-bit memory flip error. Before conducting the parity check, HP SPM itself must be initialized first.

Detailed Configuration Steps:

- Set `HP_SYSTEM_SPM_INIT_EN` and clear `HP_SYSTEM_SPM_INIT_CNT_RESET` to start initializing HP SPM.

- Read the value of HP_SYSTEM_SPM_INIT_DONE.
- Once HP_SYSTEM_SPM_INIT_DONE reaches a value of 1, set HP_SYSTEM_HP_SPM_PARITY_CHECK_EN to start parity check.
- (Recommended) clear HP_SYSTEM_SPM_INIT_EN and set HP_SYSTEM_SPM_INIT_CNT_RESET after HP_SYSTEM_SPM_INIT_CNT_RESET reaches 1 to complete the initialization process.

During the above process, avoid any access to HP SPM.

7.3.3 External Memory

ESP32-P4 supports SPI, dual SPI, quad SPI and QPI interfaces for connecting to external flash, as well as OPI and HPI interfaces for connecting to external RAM.

ESP32-P4 provides the following security features to protect users' data in external flash and RAM:

- hardware manual encryption and automatic decryption based on XTS-AES algorithm to protect users' programs and data stored in the external flash.
- automatic encryption and decryption based on XTS-AES algorithm to protect users' data in the external RAM.

7.3.3.1 External Memory Address Mapping

HP CPU can access external flash or RAM via 0x4000_0000 ~ 0x43FF_FFFF or via 0x4800_0000 ~ 0x4BFF_FFFF using cache-able or noncache-able methods. HP CPU can also access external flash or RAM directly via 0x8000_0000 ~ 0x83FF_FFFF or via 0x8800_0000 ~ 0x8BFF_FFFF, which is slower than the access via cache and is therefore commonly used for debugging.

When the HP CPU accesses external memory, it maps the addresses to physical addresses based on the information in the Memory Management Unit (MMU).

- Range (0x4000_0000 ~ 0x43FF_FFFF) is mapped to the physical address of the external flash accessed by HP CPU via cache.
- Range (0x4800_0000 ~ 0x4BFF_FFFF) is mapped to the physical address of the external RAM accessed by HP CPU via cache.
- Range (0x8000_0000 ~ 0x83FF_FFFF) is mapped to the physical address of the external flash accessed by HP CPU directly.
- Range (0x8800_0000 ~ 0x8BFF_FFFF) is mapped to the physical address of the external RAM accessed by HP CPU directly.

This mapping allows ESP32-P4 to address up to 64 MB of external flash and 64 MB of external RAM. **Note that** the instruction bus shares the same address space (64 MB) with the data bus for accessing external memory.

7.3.3.2 Cache

As shown in Figure 7.3-3, ESP32-P4 has a two-level cache system, with the following features:

- L1 cache, including:
 - 16 KB of instruction cache (icache) with a 64 B block size, four-way set associative

- 64 KB of data cache (dcache) with a 64 B block size, two-way set associative, supporting two writing strategies: write-through and write-back
- L2 cache: 128 KB/256 KB/512 KB with 64 B/128 B block sizes, eight-way set associative
- Cache-able and noncache-able access, depending on CPU PMA
- Preload operation
- Lock operation
- Critical word first and early restart

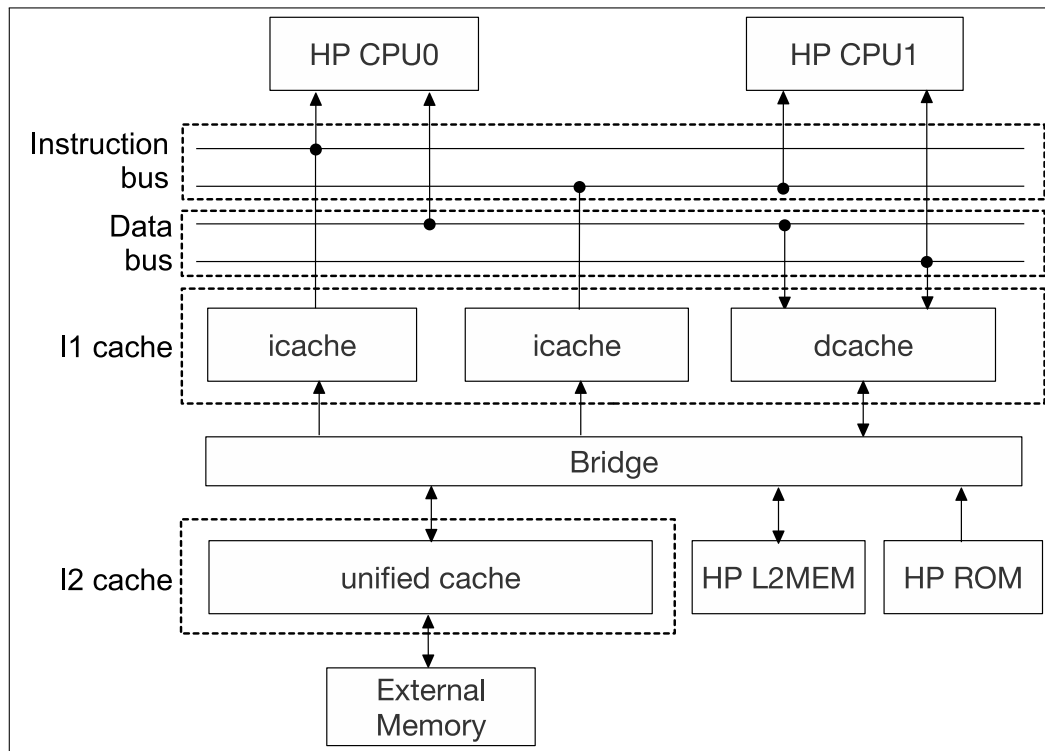


Figure 7.3-3. Cache Structure

7.3.3.3 Cache Operations

ESP32-P4 I1 cache and I2 cache supports the following operations:

1. **Write-back** (for dcache and I2 cache only):

- Clears dirty bits in the tag memory and updates new data to external memory.
- After the write-back operation, the new data is updated to external memory.
- Users can choose whether to invalidate the data in the cache.
- If the data is not invalidated, CPU will read/write data directly from/to the cache where data missing will not occur.

2. **Clean** (for dcache and I2 cache only):

- Clears dirty bits in the tag memory without updating data to external memory.
- After the clean operation, old data remains in external memory, while the cache holds the new one (unaware of it).

- CPU can then read/write the data directly from/to the cache, where data missing will not occur.

3. Invalidate:

- Removes valid data from the cache.
- Once this operation is done, the deleted data is stored only in external memory.
- HP CPU needs to access external memory if it wants to access the data again.
- Two types of invalidate operation:
 - Manual-Invalidate: is performed only on data in the specified cache area.
 - Invalidate-All: is performed on all data in the cache.

4. Preload:

- Loads instructions and data into the cache in advance.
- Minimum unit of preload-operation is one block.
- Two types of preload operation:
 - Manual-Preload: involves the hardware prefetching continuous data based on the virtual address specified by the software.
 - Auto-Preload: involves the hardware prefetching continuous data based on the current address where the cache hits or misses (configuration-dependent).

5. Lock/Unlock:

- Lock operation prevents easily replacing data in the cache.
- Two types of lock:
 - Prelock: locks data in the specified area when filling missing data to cache memory, leaving the data outside the specified area unlocked.
 - Manual Lock: checks data in the cache memory and locks data only if it falls in the specified area, leaving data outside the specified area unlocked.
- When there is missing data, the cache replaces the data in the unlocked way first, ensuring that the data in the locked way remains in the cache and will not be replaced.
- When all ways within the cache are locked, the cache will replace data, as if it was not locked.
- Unlocking is the reverse of locking and can only be done manually.
- Note: Manual-Invalidate operation only works on the unlocked data. If you plan to perform this operation on the locked data, please unlock them first.

7.3.4 DMA Address Space

ESP32-P4 includes AHB General Direct Memory Access (GDMA-AHB), AXI General Direct Memory Access (GDMA-AXI), and VDMA, all providing general direct memory access services.

In ESP32-P4, masters like USB 2.0 High-speed, USB 2.0 Full-speed, GDMAC, Trace0/1, VDMA, 2D-DMA, and H264 Encoder have built-in DMA function modules, providing direct memory access services.

All DMA-enabled master devices mentioned above can access:

- HP L2MEM (0x4FF0_0000~0x4FFB_FFFF).
- external flash (0x4000_0000 ~ 0x43FF_FFFF).
- external RAM (0x4800_0000 ~ 0x4BFF_FFFF).

Additionally, VDMA Master 1 can also access:

- MIPI CSI internal memory (0x5010_4000 ~ 0x5010_4FFF).
- MIPI DSI internal memory (0x5010_5000 ~ 0x5010_5FFF).

Note:

When accessing a memory via DMA, a corresponding access permission is needed, otherwise this access may fail. For more information about permission control, please refer to Chapter 19 [Permission Control \(PMS\)](#).

7.3.5 Modules/Peripherals Address Mapping

Table 7.3-2 lists all the modules/peripherals and their respective address ranges. Note that the address space of specific modules/peripherals is defined by “Boundary Address” (including both Low Address and High Address).

Table 7.3-2. Module/Peripheral Address Mapping

Target	Boundary Address		Size (KB)
	Low Address	High Address	
HP CPU Peripherals (HP CPU PERI)			
Reserved	0x3FF0_0000	0x3FF0_3FFF	
RISC-V Trace Encoder 0	0x3FF0_4000	0x3FF0_4FFF	4
RISC-V Trace Encoder 1	0x3FF0_5000	0x3FF0_5FFF	4
Bus Monitor	0x3FF0_6000	0x3FF0_6FFF	4
Reserved	0x3FF0_7000	0x3FF0_DFFF	
L2MEM Monitor	0x3FF0_E000	0x3FF0_EFFF	4
SPM Monitor	0x3FF0_F000	0x3FF0_FFFF	4
HP Peripherals 0 (HP PERIO)			
USB 2.0 OTG High-Speed	0x5000_0000	0x5003_FFFF	256
USB 2.0 OTG Full-Speed	0x5004_0000	0x5007_FFFF	256
USB 2.0 OTG Full-Speed PHY	0x5008_0000	0x5008_0FFF	4
VDMA Controller	0x5008_1000	0x5008_1FFF	4
Reserved	0x5008_2000	0x5008_2FFF	
SD/MMC Host Controller	0x5008_3000	0x5008_3FFF	4
H264 Encoder	0x5008_4000	0x5008_4FFF	4
GDMA-AHB	0x5008_5000	0x5008_5FFF	4
JPEG Codec	0x5008_6000	0x5008_6FFF	4
Pixel-Processing Accelerator (PPA)	0x5008_7000	0x5008_7FFF	4
2D-DMA Controller	0x5008_8000	0x5008_8FFF	4
Key Manager	0x5008_9000	0x5008_9FFF	4

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Table 7.3-2 – cont'd from previous page

Target	Boundary Address		Size (KB)
	Low Address	High Address	
GDMA-AXI	0x5008_A000	0x5008_BFFF	8
AES Accelerator (AES)	0x5009_0000	0x5009_0FFF	4
SHA Accelerator (SHA)	0x5009_1000	0x5009_1FFF	4
RSA Accelerator (RSA)	0x5009_2000	0x5009_2FFF	4
RSA Digital Signature Peripheral (RSA_DS)	0x5009_3000	0x5009_3FFF	4
HMAC Accelerator (HMAC)	0x5009_4000	0x5009_4FFF	4
ECC Accelerator (ECC)	0x5009_5000	0x5009_5FFF	4
ECDSA_DS	0x5009_6000	0x5009_6FFF	4
Reserved	0x5009_7000	0x5009_7FFF	
GMAC	0x5009_8000	0x5009_BFFF	16
USB 2.0 OTG High-Speed PHY	0x5009_C000	0x5009_CFFF	4
Reserved	0x5009_D000	0x5009_DFFF	
CSI Host	0x5009_F000	0x500F_FFFF	4
DSI Host	0x500A_0000	0x500A_0FFF	4
Image Signal Processor (ISP)	0x500A_1000	0x500A_1FFF	4
Remote Control Peripheral (RMT)	0x500A_2000	0x500A_2FFF	4
Bit-scrambler	0x500A_3000	0x500A_3FFF	4
AXI ICM	0x500A_4000	0x500A_4FFF	4
HP Peripheral Permission (HP_PERI_PMS)	0x500A_5000	0x5009_57FF	2
LP2HP Peripheral Permission (LP2HP_PERI_PMS)	0x500A_5800	0x500A_5FFF	2
HP DMA Permission (HP_DMA_PMS)	0x500A_6000	0x500A_6FFF	4
H264 DMA	0x500A_7000	0x500A_7FFF	4
Reserved	0x500A_8000	0x500B_FFFF	
HP Peripherals 1 (HP PERI1)			
Motor Control PWM0 (MCPWM0)	0x500C_0000	0x500C_0FFF	4
Motor Control PWM1 (MCPWM1)	0x500C_1000	0x500C_1FFF	4
Timer Group 0 (TIMG0)	0x500C_2000	0x500C_2FFF	4
Timer Group 1 (TIMG1)	0x500C_3000	0x500C_3FFF	4
I2C Controller0 (I2C0)	0x500C_4000	0x500C_4FFF	4
I2C Controller1 (I2C1)	0x500C_5000	0x500C_5FFF	4
I2S Controller0 (I2S0)	0x500C_6000	0x500C_6FFF	4
I2S Controller1 (I2S1)	0x500C_7000	0x500C_7FFF	4
I2S Controller0 (I2S2)	0x500C_8000	0x500C_8FFF	4
Pulse Count Controller (PCNT)	0x500C_9000	0x500C_9FFF	4
UART Controller 0 (UART0)	0x500C_A000	0x500C_AFFF	4
UART Controller 1 (UART1)	0x500C_B000	0x500C_BFFF	4
UART Controller 2 (UART2)	0x500C_C000	0x500C_CFFF	4
UART Controller 3 (UART3)	0x500C_D000	0x500C_DFFF	4
UART Controller 4 (UART4)	0x500C_E000	0x500C_EFFF	4
Parallel IO Controller (PARL_IO)	0x500C_F000	0x500C_FFFF	4

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Table 7.3-2 – cont'd from previous page

Target	Boundary Address		Size (KB)
	Low Address	High Address	
General Purpose SPI2 (GP-SPI2)	0x500D_0000	0x500D_0FFF	4
General Purpose SPI3 (GP-SPI3)	0x500D_1000	0x500D_1FFF	4
USB Serial/JTAG Controller	0x500D_2000	0x500D_2FFF	4
LED PWM Controller (LEDC)	0x500D_3000	0x500D_3FFF	4
Reserved	0x500D_4000	0x500D_4FFF	
Event Task Matrix (SOC_ETM)	0x500D_5000	0x500D_5FFF	4
Interrupt Matrix (INTMTX)	0x500D_6000	0x500D_6FFF	4
Two-wire Automotive Interface 0 (TWAIO)	0x500D_7000	0x500D_7FFF	4
Two-wire Automotive Interface 1 (TWA1)	0x500D_8000	0x500D_8FFF	4
Two-wire Automotive Interface 2 (TWA2)	0x500D_9000	0x500D_9FFF	4
I3C Master Controller	0x500D_A000	0x500D_AFFF	4
I3C Slave Controller	0x500D_B000	0x500D_BFFF	4
LCD and Camera Controller (LCD_CAM)	0x500D_C000	0x500D_CFFF	4
Reserved	0x500D_D000	0x500D_DFFF	
ADC Controller	0x500D_E000	0x500D_EFFF	4
UHCI Controller (UHCI)	0x500D_F000	0x500D_FFFF	4
GPIO Matrix	0x500E_0000	0x500E_0FFF	4
IO MUX	0x500E_1000	0x500E_1FFF	4
System Timer (SYSTIMER)	0x500E_2000	0x500E_2FFF	4
Reserved	0x500E_3000	0x500E_4FFF	
HP System Register (SYSREG)	0x500E_5000	0x500E_5FFF	4
Reset and Clock	0x500E_6000	0x500E_6FFF	4
Reserved	0x500E_7000	0x500E_FFFF	
MIPI Camera and LCD Memory			
MIPI Camera Memory	0x5010_4000	0x5010_4FFF	4
MIPI LCD Memory	0x5010_5000	0x5010_5FFF	4
LP Always On Peripherals (LP AON PERI)			
LP System Register	0x5011_0000	0x5011_0FFF	4
LP Always-on Clock and Reset	0x5011_1000	0x5011_1FFF	4
LP Timer	0x5011_2000	0x5011_2FFF	4
LP Analog peripherals (LP ANAPERI)	0x5011_3000	0x5011_3FFF	4
LP HUK	0x5011_4000	0x5011_4FFF	4
Power Management Unit (PMU)	0x5011_5000	0x5011_5FFF	4
LP Watch Dog Timer (LP WDT)	0x5011_6000	0x5011_6FFF	4
Reserved	0x5011_7000	0x5011_7FFF	
LP Mailbox (LP MB)	0x5011_8000	0x5011_8FFF	4
Reserved	0x5011_A000	0x5011_FFFF	
LP Peripherals (LP PERI)			
LP Peripheral Clock and Reset (LP PERI-CLKRST)	0x5012_0000	0x5012_0FFF	4
LP UART	0x5012_1000	0x5012_1FFF	4

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Table 7.3-2 – cont'd from previous page

Target	Boundary Address		Size (KB)
	Low Address	High Address	
LP I2C	0x5012_2000	0x5012_2FFF	4
LP SPI	0x5012_3000	0x5012_3FFF	4
LP Analog I2C	0x5012_4000	0x5012_4FFF	4
LP I2S	0x5012_5000	0x5012_5FFF	4
Reserved	0x5012_6000	0x5012_6FFF	4
LP ADC	0x5012_7000	0x5012_7FFF	4
Reserved	0x5012_8000	0x5012_9FFF	
LP GPIO Matrix	0x5012_A000	0x5012_AFFF	4
LP IO MUX	0x5012_B000	0x5012_BFFF	4
LP Interrupt (LP INTR)	0x5012_C000	0x5012_CFFF	4
LP eFuse	0x5012_D000	0x5012_DFFF	4
LP Peripheral Permission (LP_PERI_PMS)	0x5012_E000	0x5012_E7FF	2
HP2LP Peripheral Permission (HP2LP_PERI_PMS)	0x5012_E800	0x5012_EFFF	2
LP Temperature Sensor	0x5012_F000	0x5012_FFFF	4

Note:

As shown in Figure 7.3-1, HP CPU and LP CPU can access HP CPU PERI, HP PERIO, HP PERI1, LP AON PERI and LP PERI listed in Table 7.3-2.